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A COMBINED GEOCHRONOLOGICAL AND PALEOMAGNETIC STUDY ON ~1220 MA MAFIC DIKES IN THE NORTH CHINA CRATON AND THE IMPLICATIONS FOR THE BREAKUP OF NUNA AND ASSEMBLY OF RODINIA

JIKAI DING*, SHIHONG ZHANG*,†, HANQING ZHAO*, HANBIAO XIAN*, HAIYAN LI*, TIANSHUI YANG*, HUAICHUN WU*, and WEI WANG**

ABSTRACT. A new combined geochronological and paleomagnetic study was carried out on the late Mesoproterozoic mafic dikes in the northeastern and central North China Craton (NCC). Three dikes were dated at 1224.2 ± 5.1 Ma, 1219.9 ± 6.0 Ma and 1215.2 ± 7.7 Ma through use of the secondary ion mass spectrometry (SIMS) zircon dating method and one dike was dated at 1219.1 ± 4.3 Ma by using SIMS baddeleyite dating method, indicating that these dikes emplaced at ~1220 Ma. A total of 286 paleomagnetic core samples were collected from seven ~1220 Ma dikes and their host rocks, as well as two ~775 Ma dikes for field tests. All the samples were subjected to stepwise thermal demagnetization. Rock magnetic experiments of representative samples demonstrate that the dominant magnetic carriers of the ~1220 Ma dikes are single-domain and pseudo-single domain magnetite grains. Two paleomagnetic components were identified from the ~1220 Ma dikes. The low temperature component is a viscous magnetic remanence acquired in the recent geomagnetic field. The high temperature component (HTC) was isolated between 500 and 580°C, which directs east and down with moderate inclination. A paleomagnetic pole (11.3°N, 175.8°E, A_{95} = 7.6°) was determined by averaging seven virtual geomagnetic poles (VGPs), each VGP corresponding to the mean HTC direction of an individual dike. The ~1220 Ma pole passed a baked-contact test and appeared to adequately average out secular variation. It differs from any younger poles of the NCC, and is thus considered as a reliable pole for the NCC. While our new results place the NCC at intermediate paleolatitudes (~28.8°N for site located at 36.8°N, 113.4°E) at ~1220 Ma, the coeval high-quality poles in the global paleomagnetic database, however, indicate that the Australian cratons were positioned near the pole and Laurentia was located around the Equator, depicting a dispersed paleogeographic pattern of the three continents at ~1220 Ma. The updated paleomagnetic data together with geological evidence may suggest that the NCC separated from northern Australia between ~1320 and ~1220 Ma, and later joined northwestern Laurentia between ~1220 and ~1110 Ma.

Keywords: paleomagnetism, ~1220 Ma mafic dikes, North China Craton, Nuna, Rodinia

INTRODUCTION

It has been widely recognized that the past two billion years have witnessed at least three supercontinents, named Nuna (also Columbia, for name discussion see in Evans, 2013; Meert and Santosh, 2017), Rodinia and Pangea from old to young. Episodic amalgamation and dispersal of the supercontinents have profoundly influenced the evolution of Earth's system (for example, Nance and Murphy, 2013; Nance and others, 2014). The late Mesoproterozoic–early Neoproterozoic (1300–900 Ma) is generally regarded as the critical time of the breakup of Nuna and the subsequent assembly of Rodinia (Li and others, 2008; Pisarevsky and others, 2014a; Evans and others, 2016b), but the transition process between the two supercontinents remains largely unknown

* State Key Laboratory of Biogeology and Environmental Geology, China University of Geosciences, Beijing 100083, China
** State Key Laboratory of Geological Processes and Mineral Resources, China University of Geosciences, Beijing 100083, China
† Corresponding Author: Shihong Zhang, shzhang@cugb.edu.cn

(Zhang and others, 2012; Pisarevsky and others, 2014a, 2014b; Meert and Santosh, 2017). There are myriad inferences about the style and rate of plate motions during the Mesoproterozoic which remain enigmatic due to the paucity of reliable paleomagnetic data (Li and others, 2008; Zhang and others, 2012; Evans, 2013; Meert, 2014; Pisarevsky and others, 2014a, 2014b; Evans and others, 2016b; Meert and Santosh, 2017).

Recently, several ~1220 Ma mafic dikes were reported from the North China Craton (NCC) (Peng and others, 2013; Xiang, ms, 2014; Peng, 2015; Wang and others, 2015, 2016). We carried out a combined paleomagnetic and SIMS U-Pb geochronological study for seven fresh dikes, and obtained a high-quality paleomagnetic pole for the NCC at ~1220 Ma. This pole together with coeval high-quality paleomagnetic data from other continents and geological information allow us to present a paleogeographic map at ~1220 Ma, which suggests large-scale plate motion from the breakup of Nuna to the assembly of Rodinia.

Geological background and sampling

The sampling trips were carried out in three regions, named the Jianping region (figs. 1B and 1C), the Lulong region (figs. 1B and 1D) in the northeastern NCC, and the Licheng region in the central NCC (figs. 1B and 1E).

In the Jianping region, three ~1230 Ma mafic dikes were previously reported (fig. 1C; Wang and others, 2015). The dikes vertically or subvertically emplaced into the Archean–Paleoproterozoic ($Ar–Pt_1$) gneisses and the ~1.7 Ga alkaline plutons (Wang and others, 2013a, 2013b). They are generally ~20 to 50 m wide, more than ~5 km long and NNE-trending (fig. 1C). Lithologically, the dikes are diabase or gabbro, with the main mineral assemblage of plagioclase and clinopyroxene. They are mainly alkaline in composition with ocean island basalt (OIB)-type geochemical features (Wang and others, 2015). Among them, two dikes were dated at ~1230 Ma (fig. 1C; dike JP01, 1229 ± 10 Ma, 1231 ± 16 Ma; dike JP03, 1229 ± 4 Ma; Wang and others, 2015) using LA-ICP-MS zircon dating method. In this study, 119 paleomagnetic samples from these three dikes were collected. In addition, 18 samples of two sites (14JP05, 14JP13) from the gneissic host rocks were collected for a baked-contact test. In order to obtain more accurate age for the sampled dikes, coarse-grained block samples (>20 kg) from the dikes were also collected to separate zircon and baddeleyite grains for the U-Pb geochronological analyses.

In the Lulong region, two groups of the Precambrian mafic dikes were reported. They are different in age, trend, lithology and many geochemical features (fig. 1D; Wang and others, 2016). The older dikes are N-S trending. They are gabbro in lithology (figs. 2A and 2B) and tholeiitic in composition. One of the dikes was dated at 1236.4 ± 7.3 Ma (dike LL02, fig. 1D; baddeleyite, SIMS, Pb-Pb; Wang and others, 2016). The younger dikes are NEE-SWW trending. They are porphyritic diabase in lithology (figs. 2C and 2D) and alkaline in composition with OIB-type geochemical features. One of the dikes was dated at 775 ± 5 Ma (dike LL04, fig. 1D; zircon, SIMS; Wang and others, 2016). All these dikes are ~5 to 30 m wide, and vertically intruded into the $Ar–Pt_1$ gneisses. In this study, 84 paleomagnetic samples were collected from three ~1236 Ma dikes (dikes LL02, LL01 and LL03, fig. 1D) and 30 samples were collected from two ~775 Ma dikes (dikes LL04 and LL05, fig. 1D).

In the Licheng region, 35 paleomagnetic samples were collected from a 30-meter-wide diabase dike (LC01) in the location ~30 km north of the Licheng County (fig. 1E). The dike is NW-SE trending and subvertically intruded the Archean gneisses and Paleoproterozoic strata, the Changzhougou and Chuanlinggou formations, which slightly dip to the west with a dip angle of ~5°. A previous study reported a ~1220 Ma mafic dike in this region, but did not provide the exact location (Peng, 2015). To constrain the age of the dike (LC01) from which we collected paleomagnetic samples,

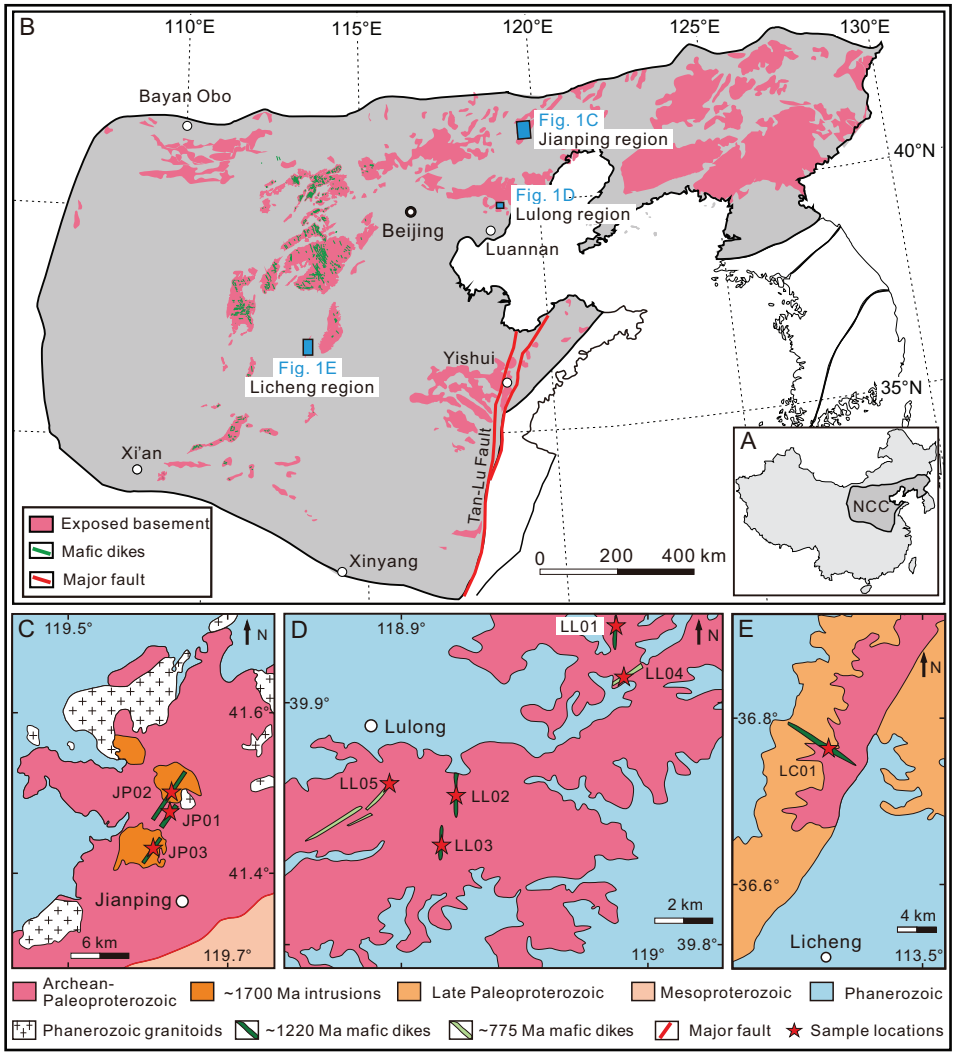


Fig. 1. (A) The location of the North China Craton (NCC). (B) Tectonic sketch map of the NCC (modified from Zhao and others, 2012) and the locations of study regions (blue rectangles). (C–E) Simplified geological maps of the (C) Jianping, (D) Lulong, and (E) Licheng regions showing the distributions of the sampled dikes.

a block sample was collected to separate zircon and baddeleyite grains for geochronological analyses.

There are more ~1220 Ma mafic dikes (intrusions) in the NCC, such as that in the Luannan region reported by Xiang (ms, 2014) and that in the Yishui region in the Tan–Lu fault zone reported by Peng and others (2013), because their attitudes and structural background are not clear yet, we did not sample them for the paleomagnetic study.

Paleomagnetic samples were collected using a portable drill, and oriented using a magnetic compass and a solar compass when possible. The difference between the results using the two orientation devices was less than 2°, indicating no significant local magnetic anomalies.

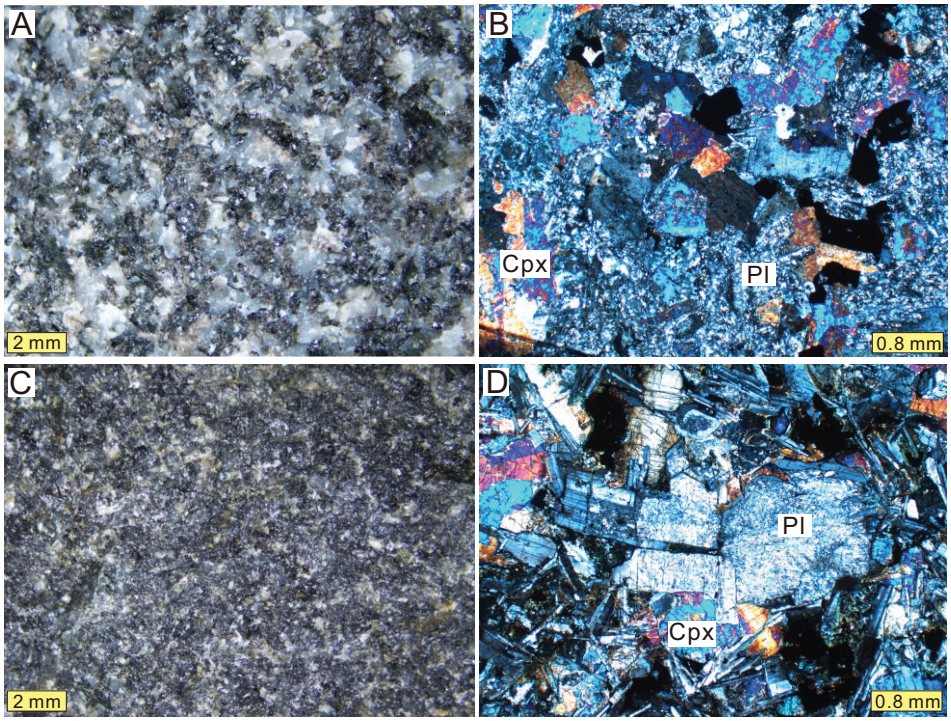


Fig. 2. Selected pictures showing occurrences and micro-textures of the ~1236 Ma mafic dikes and ~775 Ma mafic dikes in the Lulong region. (A) Gabbro and (B) its representative photomicrograph from the ~1236 Ma dike LL03. (C) Porphyritic diabase from the ~775 Ma dike LL04. (D) Porphyritic texture with large plagioclase phenocrysts in the rocks of dike LL04. Cpx, clinopyroxene; Pl, plagioclase.

GEOCHRONOLOGICAL EXPERIMENTS, ANALYSES AND RESULTS

Method and Analytical Procedure

Using conventional density and magnetic separation techniques, zircon and baddeleyite grains were separated in the mineral separation laboratory of the Bureau of Geology and Mineral Resources, Langfang, Hebei Province. Three dikes (JP01, JP02 and JP03) yielded sufficient zircon grains and one dike (LC01) yielded sufficient baddeleyite grains. The mineral grains together with the standard zircon/baddeleyite grains were mounted in the transparent epoxy. The mount was then polished to expose the longitudinal section of the crystals. All grains were documented with transmitted and reflected light micrographs, as well as cathodoluminescence (CL) images or back-scattered electron (BSE) images to reveal their internal structures.

Zircon U-Pb and baddeleyite Pb-Pb analyses were conducted on a Cameca IMS 1280-HR Large-radius ion microprobe at the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS) in Beijing, following the method of Li and others (2009, 2010). The analytical ion beams used in this study are about 20 μm × 30 μm in size. The measured U-Th-Pb ratios and their absolute abundances in the zircon grains were calibrated against the standard zircon Plešovice (Sláma and others, 2008) and 91500 (Wiedenbeck and others, 1995). The standard baddeleyite Phalaborwa (Heaman, 2009) was used to calibrate the Pb/U ratio and U concentration in the baddeleyite grains. Uncertainties on individual analyses were cited at the 1σ level, and the weighted mean ages are quoted with 95 percent confidence intervals. Data

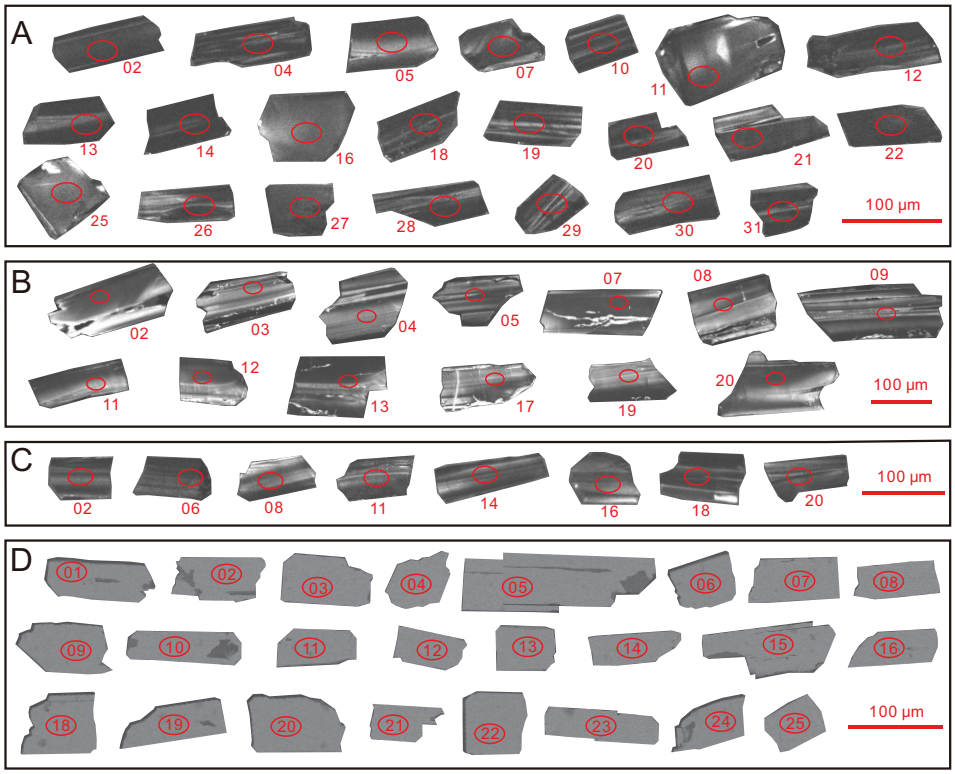


Fig. 3. (A–C) Representative CL images of the analyzed zircon grains from the ~1220 Ma dikes (A) JP01, (B) JP02 and (C) JP03 in the Jianping region. (D) Representative BSE images of the analyzed baddeleyite grains from dike LC01 in the Licheng region. Ellipses are SIMS analysis spots (20 × 30 μm).

reduction was carried out using the Isoplot/Ex v. 2.49 program (Ludwig, 2001). The results were plotted in figure 4 and the data were listed in tables A1 and A2.

Zircon U-Pb Geochronology

Three gabbro dikes in the Jianping region were dated using SIMS U-Pb zircon dating method. Most of the zircon grains are 70 to 120 μm in length and some are longer than 150 μm (figs. 3A–3C). The grains are euhedral to subhedral in morphology and they show banded or broad zoning in the CL images (figs. 3A–3C), which are the typical features of magmatic zircons from gabbro and diabase (Corfu and others, 2003).

Sample D15JP01 was collected from dike JP01. Thirty-two spots, one spot on each zircon grain, were analyzed. Some high U (>3000 ppm) grains were not used for further calculation (table A1). U and Th concentrations of the remaining zircons vary from 394 to 2893 ppm and 261 to 4857 ppm, respectively, with Th/U ratios of 0.66 to 1.77 (table A1). Twenty-five of these analyses are concordant and define a concordia age of 1224.2 ± 5.1 Ma (95% confidence, MSWD = 1.8), being identical to the weighted mean $^{207}Pb/^{206}Pb$ age of 1226.9 ± 3.4 Ma (95% confidence, MSWD = 1.9; fig. 4A). The $^{207}Pb/^{206}Pb$ dates are consistent with previous LA-ICP-MS zircon results within their uncertainties (1231 ± 16 Ma; 1231 ± 10 Ma; Wang and others, 2015). We interpreted the new concordia age of 1224.2 ± 5.1 Ma as the crystallization age of dike JP01.

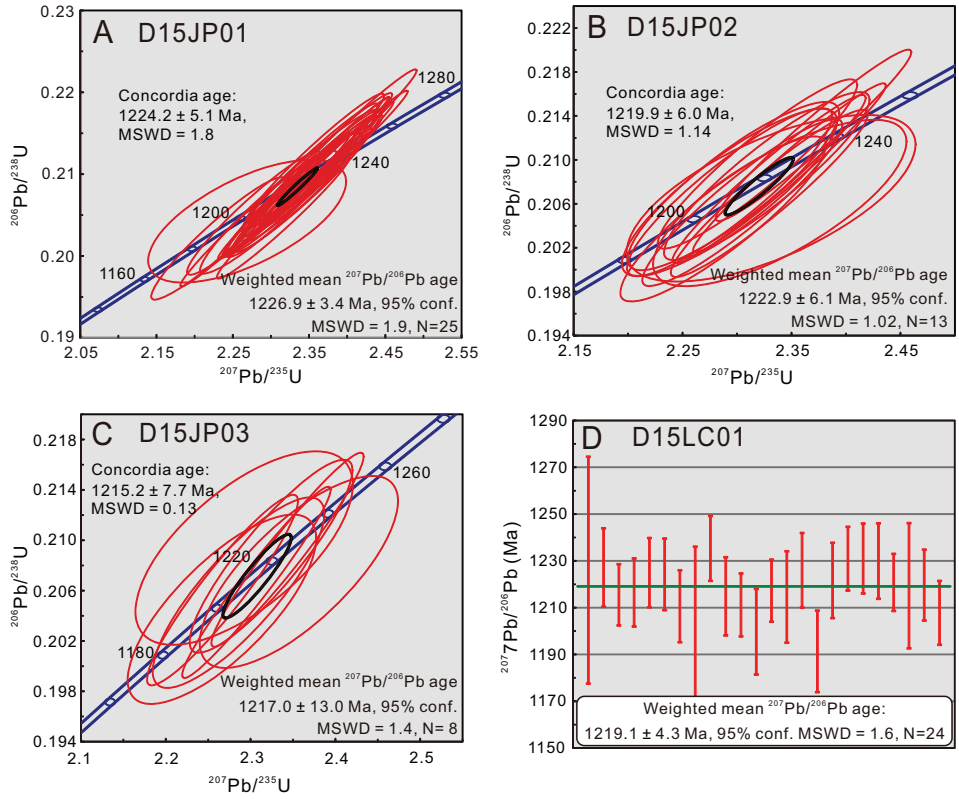


Fig. 4. (A–C) U-Pb concordia plots for zircons from the ~1220 Ma dikes in the Jianping region. (D) $^{207}Pb/^{206}Pb$ age plot for baddeleyites from dike LC01 in the Licheng region. Data-point error crosses are 2σ. MSWD, mean square of weighted deviates; conf., confidence.

Sample D15JP02 was collected from dike JP02. Twenty zircon grains were analyzed. Apart from the analyses with high common Pb or low total Pb (table A1), the remaining thirteen analyses yielded a concordia age of 1219.9 ± 6.0 Ma (95% confidence, MSWD = 1.14; fig. 4B), which is identical to the weighted mean $^{207}Pb/^{206}Pb$ age of 1222.9 ± 6.1 Ma (95% confidence, MSWD = 1.02) within their uncertainties. The Th/U ratios (0.81–1.63) and euhedral prismatic morphology indicate that the zircons are magmatic in origin. Therefore, the concordia age of 1219.9 ± 6.0 Ma is interpreted as the crystallization age of dike JP02.

Sample D15JP03 was collected from dike JP03. Twenty zircon grains were selected for analyzing. The analyses with high U, high common Pb, or low total Pb were excluded in the calculation (table A1). Eight spots define a concordia age of 1215.2 ± 7.7 Ma (95% confidence, MSWD = 0.13; fig. 4C), which is identical to the weighted mean $^{207}Pb/^{206}Pb$ age of 1217 ± 13 Ma (95% confidence, MSWD = 1.4). And the age is also consistent with previous LA-ICP-MS zircon results within their uncertainties (1229 ± 10 Ma; Wang and others, 2015). We interpreted the concordia age of 1215.2 ± 7.7 Ma as the crystallization age of dike JP03.

Baddeleyite Age

Sample D15LC01 was collected from dike LC01 in the Licheng region. The baddeleyite crystals in this sample are mostly euhedral to subhedral in shape with width of 40 to

80 μm and length of 50 to 120 μm (fig. 3D). The grains exhibit homogeneous structures in the BSE images (fig. 3D). Twenty-five baddeleyite grains were analyzed. Except for one grain with high U, the remaining grains have relatively low U content with a range of 339 to 1813 ppm (table A2). The weighted mean $^{207}Pb/^{206}Pb$ age of 24 grains is 1219.1 ± 4.3 Ma (95% confidence, MSWD = 1.2; fig. 4D), which is indistinguishable within analytical uncertainty to the previously reported results from one dike in this region (1229 ± 4 Ma; baddeleyite, SIMS, $^{207}Pb/^{206}Pb$ ages; Peng, 2015). Therefore, the $^{207}Pb/^{206}Pb$ age of 1219.1 ± 4.3 Ma is taken as the crystallization age of dike LC01.

PALEOMAGNETIC EXPERIMENTS, ANALYSES AND RESULTS

Laboratory Conditions and Procedures

Each paleomagnetic sample was cut into 1 to 3 standard cylindrical specimens of 2.5 cm in diameter and 2.2 cm in height. One specimen of each sample was subjected to stepwise thermal demagnetization, and representative specimens were subjected to rock magnetic analyses. Stepwise thermal demagnetization was conducted in intervals of 3 to 50 °C up to 580 °C using a TD-48 furnace, whose internal residual field is lower than 10 nT. Remanence was measured on a JR-6A spinner magnetometer and a 2G 755-4K superconducting magnetometer. The furnace and magnetometers are housed in a magnetically shielded room with a magnetic field below ~300 nT. Temperature-dependent low field susceptibility in an argon atmosphere was conducted using a KLY-4S Kappabridge equipped with the CS-3 high-temperature apparatus. These experiments were finished in the Paleomagnetism and Environmental Magnetism Laboratory at the China University of Geosciences, Beijing (CUGB). Hysteresis loops and back-field demagnetization of saturation isothermal remanent magnetization (IRM) were conducted using a MicroMag Model 3900 Vibrating Sample Magnetometer at the Institute of Geophysics, China Earthquake Administration. Remanence components were processed using the principal component analysis (Kirschvink, 1980), and the mean directions were calculated using Fisher statistics (Fisher, 1953). Paleomagnetic data were analyzed using the program packages developed by Enkin (ms, 1990) and Cogné (2003). Paleocontinent reconstructions were made using the GPlates freeware package (Williams and others, 2012).

Rock Magnetic Results

To identify the magnetic carriers, rock magnetic experiments were conducted on representative specimens of the ~1220 Ma dikes from the three regions. Temperature-dependent low field magnetic susceptibility curves show the Curie temperature of ~580 °C and the pronounced Hopkinson peak in the heating curve (fig. 5A), indicating the presence of PSD or SD magnetite in the specimens (Dunlop and Özdemir, 1997). Hysteresis loops and back-field demagnetization of IRM show most specimens are saturated below 500 mT with coercivities less than ~30 mT (figs. 5B–5D). On the Day diagram (Day and others, 1977; Dunlop, 2002), most hysteresis data fall in the PSD grain size area (fig. 5E). In summary, the rock magnetic results demonstrate that the main magnetic carriers in these specimens are PSD and SD magnetite grains.

Paleomagnetic Results of the Dikes

After stepwise thermal demagnetization, most samples from the seven ~1220 Ma dikes in the three regions revealed consistent results. The NRM intensities range from 0.1 to 4.0 A/m. Most specimens are characterized by two well-defined components. A low temperature component (LTC) was isolated below 300 to 400 °C (figs. 6A–6G). Its directions coincide with the geocentric axial dipole (GAD) field (figs. 7A–7G). It is thus interpreted as a viscous remanent magnetization (VRM) of the recent geomagnetic field. After removing the LTC, a high temperature component (HTC) was

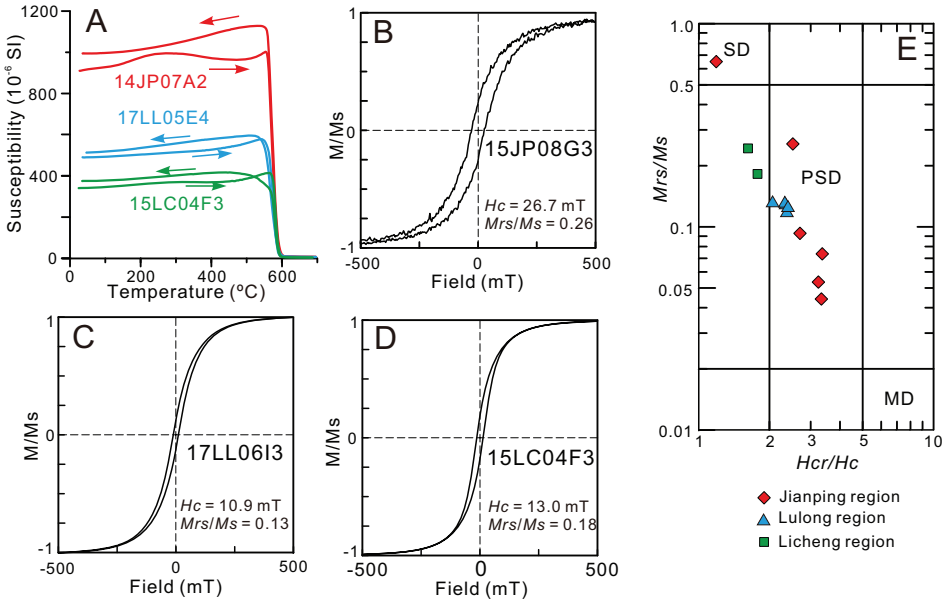


Fig. 5. Rock magnetic results for representative specimens of the ~1220 Ma dikes. (A) Temperature-dependent low field magnetic susceptibility curves. (B–D) Magnetic hysteresis loops. (E) Day plot (Day and others, 1977; Dunlop, 2002) of the hysteresis parameters. Ms, saturation magnetization; Mrs, saturation remanence; Hcr, coercivity of remanence; Hc, coercivity; SD, single-domain; PSD, pseudo-single domain; MD, multidomain.

isolated between 500 and 580 °C (figs. 6A–6G). The unblocking temperatures close to 580 °C indicate that the HTC was carried by poor-titanium magnetite, which is also supported by the rock magnetic results mentioned above. The HTC can be defined from 196 samples of the seven ~1220 Ma dikes, directing east and down with moderate inclinations (table 1; figs. 6A–6G, 8A–8F, and 8I). The dikes in the Jianping and Lulong regions vertically or subvertically intrude only the basement, indicating that they have not been tilted significantly since they emplaced. Therefore, tilt correction is not necessary for these dikes. But for dike LC01 in the Licheng region, because its host strata, the Changzhougou and Chuanlinggou formations, are slightly tilted to the west with a dip angle of 5°, we applied tilt correction to its HTC and obtained the tilt-corrected direction ($D = 89.3°$, $I = 50.9°$, $k = 154.5$, $\alpha_{95} = 2.2°$; table 1).

The corresponding virtual geomagnetic pole (VGP) of the HTC of each individual dike was calculated and listed in the table 1. By averaging the seven VGPs from the three regions (table 1; fig. 8J), we got a mean paleomagnetic pole of 11.3°N, 175.8°E ($A_{95} = 7.6°$) for the NCC at ~1220 Ma (fig. 8J; NC1220, table 2).

Most specimens of the two ~775 Ma dikes from the Lulong region also yielded two components: an LTC defined below 400 °C and an HTC defined between 500 and 580 °C. The LTC of dike LL05 directs around the local GAD field (fig. 7I) and was thus interpreted as a VRM of recent geomagnetic field, while the LTC of dike LL04 shows scattered directions (fig. 7H). The HTCs of the two dikes consistently direct ENE with shallow inclinations (figs. 8G–8H; table 1), and differ significantly from those of the ~1220 Ma dikes (figs. 8A–8F, 8I). The different HTCs observed from the ~1220 Ma and ~775 Ma dikes in the same sampling area may exclude any regional pervasive remagnetization event after ~775 Ma. Due to the limited sampling of the ~775 Ma dikes, the corresponding VGPs are unlikely to be enough to average out the secular

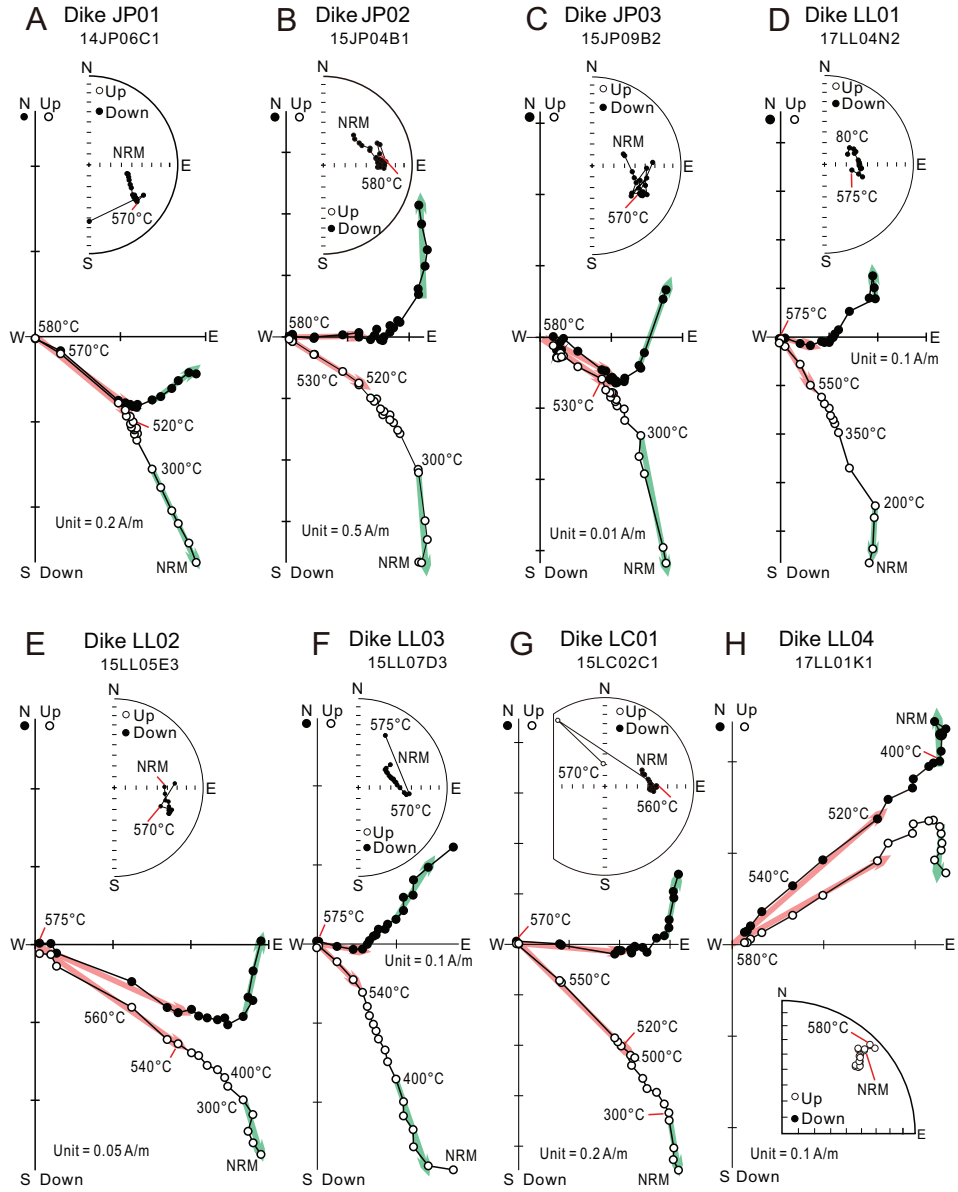


Fig. 6. Orthogonal demagnetization diagrams and equal-area projections of representative specimens from (A–G) the ~1220 Ma dikes and (H) the ~775 Ma dike. Least-squares fits are shown on the orthogonal demagnetization diagrams for the low temperature (green) and high temperature (pink) components of the demagnetization data. All are plotted in geographic coordinates. NRM, natural remanent magnetization.

variation of the geomagnetic field. They will not be used for further paleogeographic interpretation in this paper.

Baked-contact Test and Magnetization Timing

The primary nature of the HTC of the ~1220 Ma dikes in the Jianping region is supported by a positive baked-contact test. Two sites were collected from the gneissic

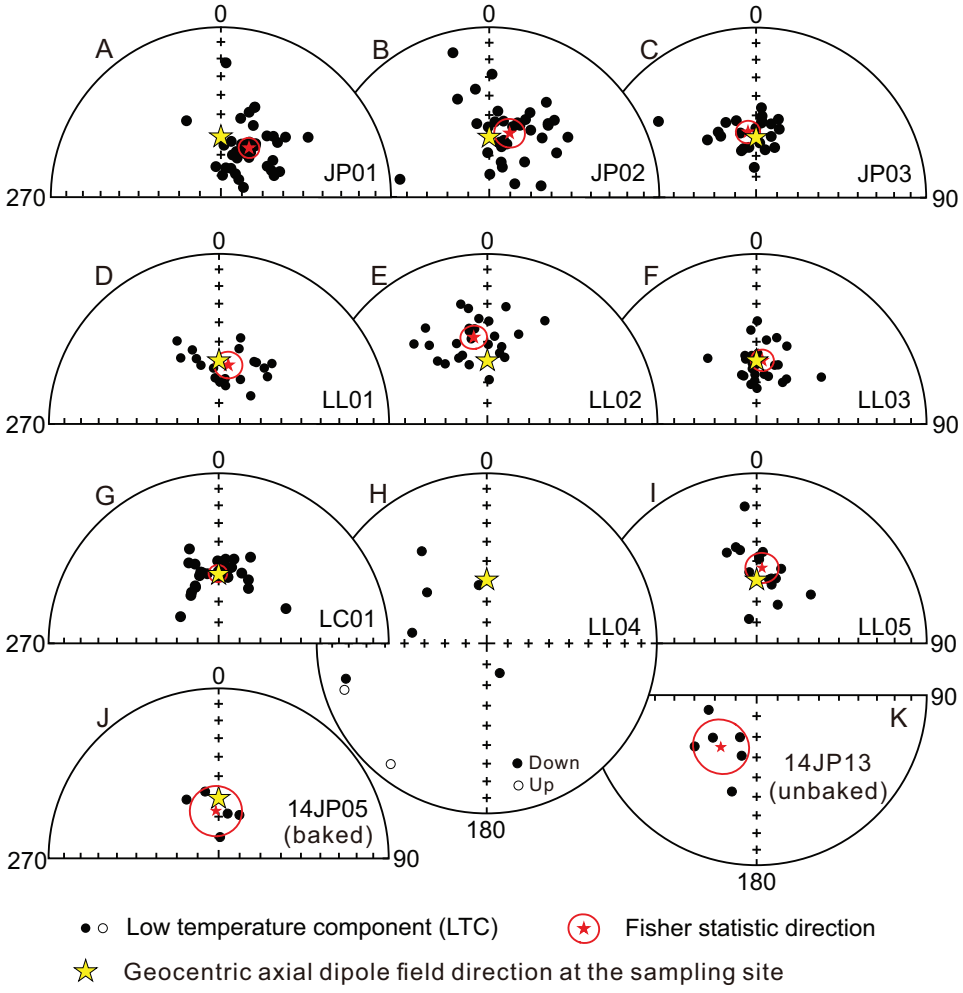


Fig. 7. Equal-area projections of the low temperature component (LTC) from (A–G) the ~1220 Ma dikes, (H–I) two ~775 Ma dikes, (J) baked and (K) unbaked host rocks. The plots are all shown in geographic coordinates.

host rocks of dike JP01 (~50 m wide). Site 14JP05 was derived from the baked host rocks at a distance of 1.8 to 4.5 m from the contact, whereas site 14JP13 from the unbaked host rocks situated about one kilometer from the contact (fig. 9A). The NRM intensities of the specimens of the host rock are 1.0 to 4.5 A/m. The two sites yielded different results after stepwise thermal demagnetization. The baked specimens (site 14JP05) recorded an LTC of recent origin (figs. 7J and 9C) and an HTC with a mean of $D = 112.9°$, $I = 57.4°$, $k = 79.9$, $\alpha_{95} = 6.8°$, resembling that of dike JP01 (figs. 9C and 9D). While the unbaked specimens (site 14JP13) contained an LTC directing SW-down with moderate inclination (figs. 7K and 9B) and an HTC with a mean of $D = 69.4°$, $I = 8.7°$, $k = 82.8$, $\alpha_{95} = 6.1°$ (figs. 9B and 9D), which differs significantly from the HTC of dike JP01. These results show that dike JP01 and the baked rocks share similar HTCs, being obviously distinct from that of the unbaked rocks. Such a positive baked-contact test implies the HTC of the ~1220 Ma dikes is a primary thermal remanent magnetization.

TABLE 1

Paleomagnetic results from the ~1220 Ma, ~775 Ma mafic dikes and the host rocks

ID	Sites	Age (Ma)	Strike (°)/Dip (°)	n/n$_0$	Slat (°N)	Slong (°E)	Dg (°)	Ig (°)	Ds (°)	Is (°)	*k*	α_{95} (°)	Plat (°N)	Plong (°E)	*D*p (°)	*D*m (°)
Jianping region, northeastern NCC, ~1220 Ma mafic dikes																
JP01	14JP06-11	1224.2 ± 5.1[a], SZ; 1229 ± 10[b], LZ; 1231 ± 16[b], LZ	125/75	39/49	41.485	119.617	97.8	52.7	97.8	52.7	20.5	5.2	16.2	179.2	4.9	7.2
JP02	15JP01-07	1219.9 ± 6.0[a], SZ	120/87	29/47	41.486	119.614	100.6	47.9	100.6	47.9	55.6	3.6	11.5	181.0	3.1	4.7
JP03	15JP08-10	1215.2 ± 7.7[a], SZ; 1229 ± 4[b], LZ	110/72	22/23	41.450	119.597	118.6	49.0	118.6	49.0	32.2	5.5	1.1	169.2	4.8	7.3
Lulong region, northeastern NCC, ~1220 Ma mafic dikes																
LL01	17LL03-04	-	98/84	29/31	39.931	118.984	112.3	55.7	112.3	55.7	63.0	3.4	8.3	167.9	3.5	4.9
LL02	17LL05-06	1236.4 ± 7.3[c], SB	97/82	24/26	39.858	118.920	114.6	49.2	114.6	49.2	52.5	4.1	2.6	170.9	3.6	5.4
LL03	17LL07-08	-	95/88	25/27	39.848	118.911	87.9	49.1	87.9	49.1	67.0	3.6	20.2	186.1	3.1	4.8
Licheng region, central NCC, ~1220 Ma mafic dikes																
LC01	15LC01-04	1219.1 ± 4.3[a], SB	30/81	28/35	36.784	113.405	87.7	46.1	89.3	50.9	154.5	2.2	18.8	177.5	2.0	3.0
Mean pole of the seven ~1220 Ma dikes													**11.3**	**175.8**	A_{95} **= 7.6**	
Lulong region, northeastern NCC, ~775 Ma mafic dikes																
LL04	17LL10	774.9 ± 5.1[c], SZ	155/84	7/10	39.912	118.994	59.5	1.6	59.5	1.6	50.3	8.6	23.5	229.1	4.3	8.6
LL05	17LL01-02	-	145/86	15/20	39.864	118.893	51.8	-27	51.8	-27	81.7	4.3	17.6	245.9	2.5	4.7
Baked-contact test: host rocks of the dike JP01																
Baked	14JP05			7/8	41.485	119.617	112.9	57.4	112.9	57.4	79.9	6.8	10.3	167.1	7.3	9.9
Unbaked	14JP13			8/10	41.488	119.626	69.4	8.7	69.4	8.7	82.8	6.1	18.3	220.3	3.1	6.2

Note: Dating method of age: SZ = SIMS zircon; SB = SIMS baddeleyite; LZ = LA-ICP-MS zircon. n/n$_0$, number of samples for statistic/number of samples for demagnetization; Slat/Slong, sample locality latitude/longitude; Dg/Ig, declination/inclination in the geographic coordinates; Ds/Is, declination/inclination in the stratigraphic coordinates; *k*, fisher precision parameter for direction; α_{95}/A$_{95}$, radius of circle of 95% confidence about the direction/pole; Plat/Plong, Latitude/Longitude of VGP in the stratigraphic coordinates. *D*m/*D*p, semi-axes of elliptical error around the pole at a probability of 95%.

[a] Age from this study.

[b] Age from Wang and others (2015).

[c] Age from Wang and others (2016).

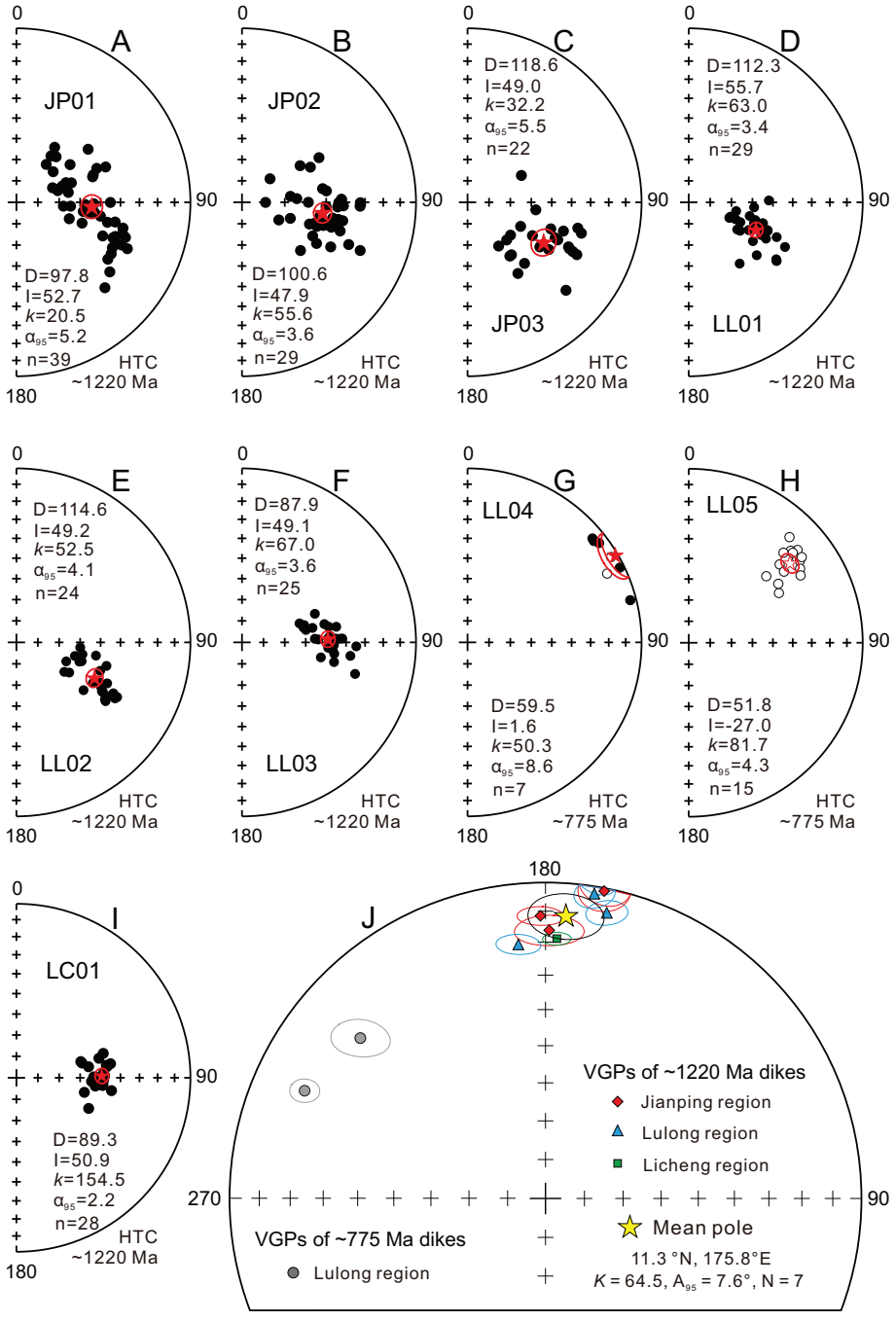


Fig. 8. (A–I) Equal-area projections of the samples' HTC directions of each dike of ~1220 Ma and ~775 Ma. Closed (open) symbols represent positive (negative) inclinations. The star indicates the mean direction of each dike. (J) Equal-area projection of virtual geomagnetic poles (VGPs) of the ~1220 Ma and ~775 Ma dikes.

TABLE 2

Paleomagnetic poles selected for the reconstructions at ~1450 Ma, ~1220 Ma, and ~930 Ma

Code	Rock unit	Age (Ma)	Age references	Plat (°)	Plon (°)	A_{95} (°)	1234567 (Q)	Paleomagnetic references
Australia								
GN	Gnowangerup dikes	1218–1203	Wingate and others (2000); Pisarevsky and others (2014b)	55.8	323.9	6.5	1111111 (7)	Pisarevsky and others (2014b)
Baltica								
LA	Lake Ladoga mafic rocks	1452–1459	Rämö and others (2001, 2005); Lubnina and others (2010)	15.2	177.1	5.5	1111111 (7)	Salminen and Pesonen (2007); Lubnina and others (2010)
B940	Mean 951–935 Ma Baltica pole	951–935	Elming and others (2014); Gong and others (2018)	−2.6	239.6	5.8	1111111 (7)	Gong and others (2018)
Laurentia								
SF	St.Francois Mtns igneous province	1476 ± 16	Meert and Stuckey (2002)	−13.2	219.0	6.1	1111101 (6)	Meert and Stuckey (2002)
MA	Michikamau anorthosite pluton	1460 ± 5	Krogh and Davies (1973)	−1.5	217.5	4.7	1111011 (6)	Emslie and others (1976)
SuD	Sudbury dikes	1235 +7/−3	Dudás and others (1994)	−2.5	192.8	2.5	1111101 (6)	Palmer and others (1977)
North China Craton								
TL	Tieling Formation	1437 ± 21; 1439 ± 14	Su and others (2010); Li and others (2014)	11.6	187.1	6.3	1111101 (6)	Wu (2005)
NC1220	~1220 Ma dikes	1236–1215	This study; Wang and others (2015, 2016)	11.3	175.8	7.6	1111101 (6)	This study
NC930	~945–920 Ma sills	945–920	Zhao and others (2019)	−28.2	141.9	10.4	1111110 (6)	Zhao and others (2019)
WS	Wangshan Formation	950–890	Wang and others (2012); He and others (2017)	26.1	320.3	5.2	1111100 (5)	Fu and others (2015)
Siberia								
OL	Olenëk mafic intrusions	1473 ± 24	Wingate and others (2009)	−33.6	73.1	10.4	1111100 (5)	Wingate and others (2009)

Note: Code, paleopoles abbreviation used in the text and reconstructions; Plat/Plon, latitude/longitude of the pole; A_{95}, the radius of circle of 95% confidence about the pole; the seven quality criteria and "Q" factor were described by Van der Voo (1990).

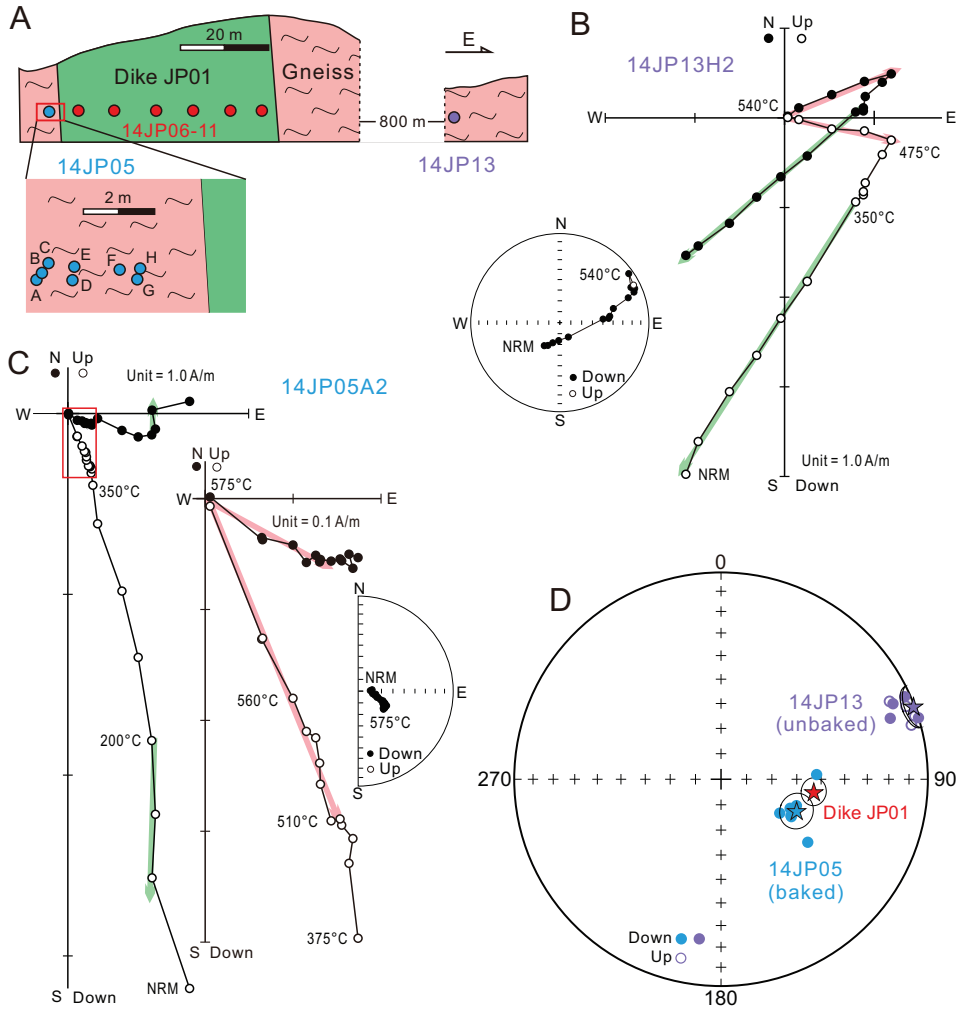


Fig. 9. Baked-contact test between the dike JP01 and its host rocks. (A) The distributions of the sampling sites from dike JP01, the baked (14JP05) and unbaked host rocks (14JP13). The inset shows the sample distributions of site 14JP05. (B–C) Orthogonal demagnetization diagrams and equal-area projections of representative specimens from the (B) unbaked and (C) baked host rocks. (D) Equal-area projection of the HTCs directions of the baked and unbaked host specimens, and the mean HTC direction of dike JP01. The stars with 95% confidence circles show the mean directions of the HTCs from the baked (blue), unbaked host rocks (purle) and dike JP01 (red). The plots in (B–D) are all shown in geographic coordinates.

Evaluation of Palaeosecular Variations

Paleomagnetic data from a single dike represents only a spot reading of the geomagnetic field (Tauxe, 1993), so it is important to evaluate whether our data have averaged out the palaeosecular variations (PSVs). In this study, two methods were applied to ascertain whether the results have sufficiently well-sampled PSV. The first method is comparing the angular dispersion (S) of VGPs of a sample set with PSV models (Biggin and others, 2008; Smirnov and others, 2011; Veikkolainen and Pesonen, 2014). The S of VGPs was calculated using the method of Biggin and others (2008), and the confidence interval (1σ) was calculated using the n−1 jackknife method (Efron, 1982). The S of the seven VGPs is 9.7°, with a 95 percent confidence interval of (8.0°, 11.4°). This result has an overlap with the

expected value (10.2°, 15.2°) for paleolatitude of 28.8° (reference point: 36.8°N, 113.4°E) according to the PSV model for 1.0 to 2.2 Ga (Smirnov and others, 2011). While it is a little smaller than the expected value (11.4°, 13.7°) according to the PSV model for 0.5 to 1.5 Ga (Veikkolainen and Pesonen, 2014). These comparable results indicate the secular variation appears to have averaged out. The other method is the N-dependent A_{95} envelope (Deenen and others, 2014), which bounded an upper limit A_{95max} and a lower limit A_{95min} to evaluate whether or not a distribution has sufficiently well-sampled PSV (Deenen and others, 2014). The A_{95} of our pole is 7.6°, which is within the A_{95} envelope ($A_{95min} = 5.5°$, $A_{95max} = 24.1°$), suggesting the PSV could have averaged. In summary, the ~1220 Ma pole of the NCC (pole NC1220, table 2) we report here has a positive baked-contact test, has averaged out secular variation and does not resemble any younger poles from the NCC. Accordingly, the pole is given a Q-score (Van der Voo, 1990) of 6 because of lacking reversals.

DISCUSSION

In the global paleomagnetic database, there are two high-quality ~1220 Ma poles from other cratons. One was obtained from the Gnowangerup-Fraser dikes of Australia (pole GN, table 2). The pole passed a baked-contact test and a reversal test and was dated at 1218 to 1203 Ma using SHRIMP U-Pb zircon and baddeleyite method (Pisarevsky and others, 2014b and references therein). The other pole was obtained from the Sudbury dikes of Laurentia (pole SuD, table 2; Palmer and others, 1977), which was dated at 1235 $+7/-3$ Ma (Baddeleyite, TIMS, Pb-Pb; Dudás and others, 1994), and the pole passed a baked-contact test. The three poles (NC1220, GN and SuD listed in table 2) can be regarded as reliable poles to constrain the paleogeographic positions of these paleocontinents.

There are also several less reliable and problematic poles that need brief discussions: (1) The paleomagnetic pole obtained from the Yunmengshan Formation of the Ruyang Group in the southern margin of the NCC passed a fold test and a reversal test. It was regarded as a high-quality pole of ~1260 Ma (Zhang and others, 2006). Recently, ages obtained from the tuff beds in the Luoyukou Formation (1611 ± 8 Ma, 1634 ± 10 Ma, 1638 ± 9 Ma; zircon U-Pb LA-MC-ICPMS) in the western Henan Province (Su and others, 2012; Li and others, 2017), however, suggest that the poles obtained from the Luoyu and Ruyang groups in the region (Zhang and others, 2000, 2006), which once were considered as of 1.3 to 0.8 Ga, should be of ~1.78 to 1.61 Ga. (2) A pole was obtained from the igneous rocks of the Nova Floresta Formation in the Amazonia Craton. Although the rocks have well-determined age (~1200 Ma, ^{40}Ar-^{39}Ar, biotite; Tohver and others, 2002), the pole does not hold any positive field test to support the interpretation of its primary origin. (3) The pole was obtained from the Late Kibaran intrusions of Congo (Meert and others, 1994). Its age was originally placed at 1236 ± 24 Ma (Meert and others, 1994), but recent work indicates the pole may be ~100 million years older than it was estimated (Maier and others, 2007; Tack and others, 2010). (4) The pole was obtained from the Harohalli alkaline dikes of India. The paleomagnetic data had dual polarity which failed reversal test (Radhakrishna and Mathew, 1996; Pradhan and others, 2008). It could average out the PSV based on the N-dependent A_{95} envelope method (Deenen and others, 2014) or comparing the S of VGPs with PSV models (Smirnov and others, 2011; Veikkolainen and Pesonen, 2014). But the lack of convincing field tests makes this pole less reliable. These problematic poles mentioned above will not be further considered in the paleogeographic reconstructions in this study.

The three reliable poles of ~1220 Ma listed in the table 2 (poles GN, SuD and NC1220) place the Australian cratons in the near-pole region (fig. 10A; Pisarevsky and others, 2014b), Laurentia in the equatorial region with its present western margin facing north (table 2; fig. 10A; Palmer and others, 1977; Dudás and others, 1994), and the NCC in the moderate paleolatitudes with its present northern margin facing west

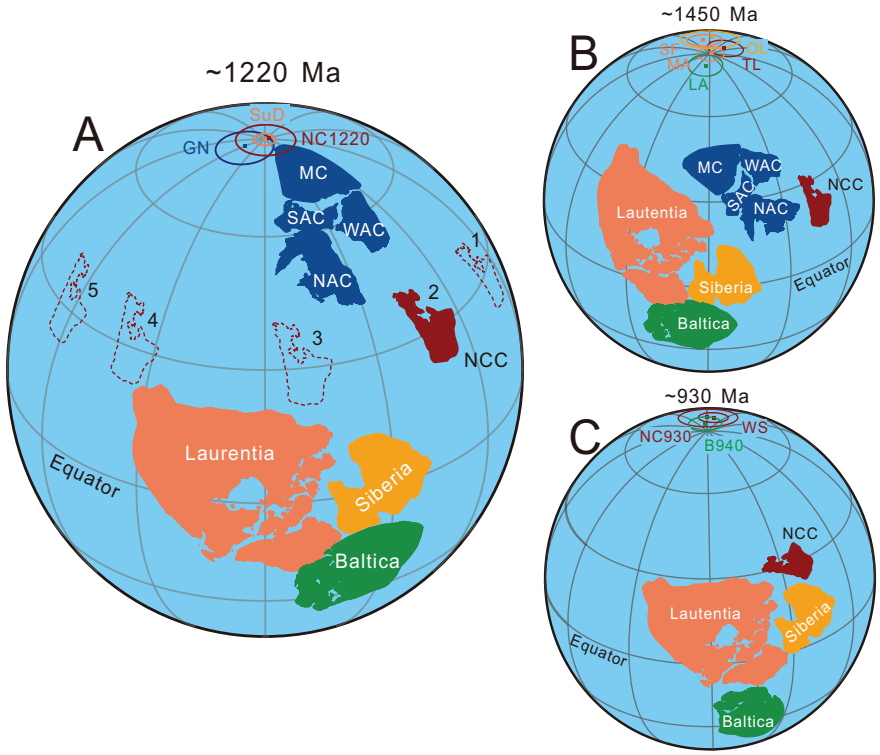


Fig. 10. Paleogeographic reconstructions of the NCC, Australian cratons, Laurentia, Baltica and Siberia (consists of Aldan and Anabar) at ~1450 Ma, ~1220 Ma and ~930 Ma. Continents and their poles are matched in color. Paleopoles used are listed in the table 2. (A) ~1220 Ma reconstruction. The Australian cratons, Laurentia and NCC (NCC1–5) are reconstructed based on their ~1220 Ma poles "GN", "SuD" and "NC1220", respectively. Baltica-Laurentia connection follows that in Nuna (Cawood and others, 2010) as shown in (B). The long-lived "tight connection" model of Siberia and Laurentia (Ernst and others, 2016; Evans and others, 2016a) was adopted in all three reconstructions. The dashed outlines of the NCC show the different options because of the fromdom of paleolongitude. (B) ~1450 Ma reconstruction (modified after Zhang and others, 2012), and paleomagnetic poles ("SF" and "MA" in table 2) were used to plot the paleolatitude of the supercontinent. (C) ~930 Ma reconstruction (modified after Zhao and others, 2019). Baltica is reconstructed according to its ~940 Ma pole "B940", Laurentia is rotated to Baltica according to the "right-way-up" model (Cawood and Pisarevsky, 2006), and the NCC is rotated to Baltica by matching their early Neoproterozic poles (Fu and others, 2015). Euler rotation parameters are listed in table A3. MC, Mawson Craton; NAC, North Australian Craton; SAC, South Australian Craton; WAC, West Australian Craton.

(fig. 10A), respectively. Comparing the three continents' paleogeographic positions at ~1220 Ma with those in Nuna at ~1450 Ma reveals three obvious changes: (1) the NCC-northern Australia connection in Nuna (fig. 10B; for example, Zhang and others, 2012; Pisarevsky and others, 2014a) was not viable at ~1220 Ma (fig. 10A); (2) the Australia-Laurentia connection in Nuna (fig. 10B) supported by the Paleo–Mesoproterozoic paleomagnetic and geological data (Zhang and others, 2012; Pisarevsky and others, 2014a; Nordsvan and others, 2018; Pourteau and others, 2018) was not viable at ~1220 Ma, while the two cratons were separated by a large latitudinal gap at ~1220 Ma (fig. 10A; Pisarevsky and others, 2014b); (3) from ~1450 to ~1220 Ma, Laurentia experienced a ~30° counterclockwise rotation (figs. 10A and 10B), whereas the NCC experienced little rotation, indicating their independent plate motion between ~1450 and ~1220 Ma. These changes collectively indicate that Nuna had broken up by ~1220 Ma, although the global paleomagnetic database cannot constrain more precisely yet when Nuna broke up between ~1400 and ~1220 Ma.

It may be worth to mention that, due to the polarity uncertainty, the NCC could be alternatively placed in the southern hemisphere. If that, the distinct movement relative to other continents can also demonstrate that supercontinent Nuna had broken up by ~1220 Ma. Combined with NCC's position in the Nuna models (Zhang and others, 2012; Pisarevsky and others, 2014a; Meert and Santosh, 2017), given the paleolongitudinal freedom allowed, the position of NCC2 (NCC1–5 in fig. 10A) could be the final choice to minimize its relative movements from ~1450 to ~1220 Ma (figs. 10A and 10B).

Several studies seem to have reached a consensus that a long-lived connection between the NCC and northern Australia might exist between ~1780 and ~1320 Ma (for example, Zhang and others, 2012, 2017; Kirscher and others, 2018; Wang and others, 2019). Major lines of evidence include the paleomagnetic consistency (Zhang and others, 2012; Kirscher and others, 2018), the similarities of the ~1320 Ma large igneous provinces (LIPs) and sedimentary successions (Zhang and others, 2017), as well as some deposits (Wang and others, 2019) preserved in the Yanliao rift basin of the NCC and the McArthur basin of the northern Australia. If the long-lived connection really existed, the NCC did separate from the northern Australia likely between ~1320 and ~1220 Ma (fig. 10A).

The new ~1220 Ma paleomagnetic results provide important constraints for the formation of Rodinia and the relationship between the NCC and Laurentia. A recent reconstruction placed the NCC along the present northwestern margin of Laurentia from ~1110 to 890 Ma based on paleomagnetic data and detrital zircon provenance analyses (fig. 10C; Fu and others, 2015; Zhao and others, 2019), which means the NCC joined the NW Laurentia and the Yenisei side of the Siberian Craton (see the discussion in Zhao and others, 2019), by ~1110 Ma. The NCC would require at least a ~60° counterclockwise rotation relative to Laurentia during ~1220 and ~1110 Ma to close the spatial gap between the NCC and NW Laurentia. Geological evidence of this convergence process, however, has not been found yet, which could be attributed to the poorly understanding of the three regions: the northern margin of the NCC, western margin of the Siberian Craton and NW Laurentia. Our analyses predict that there were considerable geological records that were associated with the breakup of Nuna and assembly of Rodinia in these regions had been dispersed or intensively superimposed by later tectonic activities that needs to be verified.

CONCLUSIONS

Three new SIMS U-Pb zircon ages and one SIMS Pb-Pb baddeleyite age confirm that the ~1220 Ma mafic dikes developed widely in the northeastern and central parts of the NCC. A new paleomagnetic pole (11.3°N, 175.8°E, A_{95} = 7.6°) was obtained from seven ~1220 Ma mafic dikes. The pole passed a positive baked-contact test and appeared to average out secular variations. It places the NCC in a moderate paleolatitudinal region at 1220 Ma, while the coeval data from Laurentia and Australia indicate a dispersed paleogeographic relationship between the NCC and those two cratons. The updated paleomagnetic database together with the geological evidence suggest that the NCC separated from the northern Australia likely between ~1320 and ~1220 Ma and joined the NW Laurentia between ~1220 and ~1110 Ma.

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APPENDIX TABLES

TABLE A1

Cameca SIMS zircon U–Pb isotopic analyses for the mafic dikes from the Jianping region

Spot	U (ppm)	Th (ppm)	Th/U	$^{206}Pb/^{204}Pb$	Pb (ppm)	f^{206} (%)	$^{207}Pb/^{206}Pb$	±1σ (%)	$^{207}Pb/^{235}U$	±1σ (%)	$^{206}Pb/^{238}U$	±1σ (%)	$^{207}Pb/^{206}Pb$ Age (Ma)	±1σ	$^{207}Pb/^{235}U$ Age (Ma)	± 1σ	$^{206}Pb/^{238}U$ Age (Ma)	±1σ
Sample D15JP01																		
D15JP01-01[a]	4992	8019	1.61	200391	1734.85	{0.01}	0.08148	0.20	2.46909	1.51	0.2198	1.50	1233.0	3.9	1263.0	11.0	1280.7	17.4
D15JP01-02	1491	1761	1.18	35883	447.58	{0.01}	0.08128	0.40	2.32763	1.56	0.2077	1.51	1228.2	7.9	1220.8	11.1	1216.5	16.7
D15JP01-03[a]	5038	10522	2.09	79594	1899.71	{0.02}	0.08145	0.20	2.42447	1.52	0.2159	1.50	1232.4	3.9	1249.9	10.9	1260.0	17.2
D15JP01-04	1599	2022	1.26	62327	491.36	{0.02}	0.08145	0.45	2.33031	1.57	0.2075	1.50	1232.3	8.9	1221.6	11.2	1215.5	16.7
D15JP01-05	524	419	0.80	5564	143.50	{0.03}	0.08111	0.54	2.29664	1.60	0.2054	1.50	1224.0	10.7	1211.3	11.4	1204.1	16.5
D15JP01-06[a]	3323	6046	1.82	8558	327.52	{0.02}	0.08009	0.32	2.22571	1.53	0.2016	1.50	1199.1	6.3	1189.2	10.8	1183.7	16.2
D15JP01-07	1067	1476	1.38	660	1104.93	{0.10}	0.08056	1.79	2.26885	2.34	0.2043	1.50	1210.8	34.9	1202.7	16.6	1198.2	16.4
D15JP01-08[a]	7098	13864	1.95	18936	2584.50	{0.10}	0.08186	0.12	2.40587	1.51	0.2131	1.50	1242.3	2.3	1244.4	10.9	1245.6	17.0
D15JP01-09	1512	1595	1.05	136743	454.62	{0.01}	0.08118	0.26	2.36829	1.52	0.2116	1.50	1225.9	5.1	1233.1	10.9	1237.2	16.9
D15JP01-10	1467	1871	1.28	55834	451.96	{0.04}	0.08130	0.26	2.34931	1.53	0.2096	1.50	1228.7	5.2	1227.3	10.9	1226.6	16.8
D15JP01-11	1136	1444	1.27	120056	347.28	{0.01}	0.08125	0.29	2.33937	1.54	0.2088	1.51	1227.5	5.7	1224.3	11.0	1222.5	16.9
D15JP01-12	1530	1697	1.11	131464	466.70	{0.01}	0.08103	0.33	2.36982	1.54	0.2121	1.50	1222.1	6.5	1233.5	11.0	1240.1	17.0
D15JP01-13	1287	1553	1.21	38702	385.76	{0.03}	0.08205	0.33	2.31634	1.54	0.2048	1.50	1246.7	6.5	1217.3	11.0	1200.8	16.5
D15JP01-14	2893	4364	1.51	35347	932.51	{0.02}	0.08129	0.19	2.32798	1.51	0.2077	1.50	1228.4	3.7	1220.9	10.8	1216.6	16.7
D15JP01-15[a]	3373	5071	1.50	23012	1102.30	{0.08}	0.08172	0.19	2.38083	1.52	0.2113	1.50	1238.9	3.7	1236.9	10.9	1235.7	16.9
D15JP01-16	606	523	0.86	73417	169.38	{0.07}	0.08159	0.41	2.34222	1.56	0.2082	1.51	1235.8	8.1	1225.2	11.2	1219.2	16.7
D15JP01-17[a]	6100	19384	3.18	6132	2705.48	{0.31}	0.08315	0.24	2.48160	1.52	0.2165	1.50	1272.7	4.7	1266.7	11.1	1263.1	17.3
D15JP01-18	1452	1178	0.81	14477	394.12	{0.12}	0.08066	0.29	2.26593	1.53	0.2037	1.50	1213.1	5.6	1201.8	10.8	1195.4	16.4
D15JP01-19	904	759	0.84	3115	246.05	{0.02}	0.08003	0.54	2.23068	1.61	0.2021	1.51	1197.8	10.6	1190.7	11.3	1186.8	16.4
D15JP01-20	1671	1521	0.91	15528	467.48	{0.03}	0.08050	0.49	2.27806	1.58	0.2052	1.50	1209.2	9.6	1205.5	11.2	1203.4	16.5
D15JP01-21	1734	2851	1.64	26776	581.63	{0.07}	0.08163	0.24	2.39063	1.52	0.2124	1.50	1236.5	4.6	1239.8	10.9	1241.7	17.0
D15JP01-22	2648	3751	1.42	121327	858.76	{0.02}	0.08104	0.29	2.40116	1.53	0.2149	1.50	1222.4	5.6	1242.9	11.0	1254.9	17.2
D15JP01-23	394	261	0.66	14675	106.57	{0.13}	0.08120	0.50	2.35316	1.58	0.2102	1.50	1226.3	9.8	1228.5	11.3	1229.8	16.8
D15JP01-24[a]	3318	5715	1.72	62801	1148.04	{0.03}	0.08143	0.23	2.42288	1.57	0.2158	1.56	1231.8	4.6	1249.4	11.4	1259.7	17.8
D15JP01-25	745	1037	1.39	91191	236.16	{0.02}	0.08086	0.36	2.35120	1.54	0.2109	1.50	1218.0	7.0	1227.9	11.0	1233.6	16.9
D15JP01-26	2744	4857	1.77	52299	939.94	{0.04}	0.08111	0.19	2.36679	1.52	0.2116	1.51	1224.1	3.7	1232.6	10.9	1237.5	17.0
D15JP01-27	1146	2024	1.77	27500	388.87	{0.07}	0.08127	0.34	2.35457	1.54	0.2101	1.50	1227.9	6.7	1228.9	11.0	1229.5	16.8
D15JP01-28	2333	4266	1.83	25797	789.69	{0.05}	0.08108	0.25	2.31706	1.52	0.2073	1.50	1223.3	4.8	1217.5	10.8	1214.3	16.6
D15JP01-29	1099	650	0.59	15550	290.23	{0.12}	0.08162	0.30	2.35814	1.53	0.2096	1.50	1236.3	5.9	1230.0	11.0	1226.4	16.8
D15JP01-30	652	535	0.82	39819	183.77	{0.05}	0.08069	0.38	2.35822	1.57	0.2120	1.52	1214.0	7.5	1230.0	11.2	1239.2	17.2
D15JP01-31	1197	1398	1.17	23353	359.25	{0.60}	0.08121	0.30	2.32271	1.53	0.2074	1.50	1226.5	5.9	1219.3	10.9	1215.2	16.6
D15JP01-32	1877	2551	1.36	63902	593.63	{0.03}	0.08151	0.23	2.37251	1.52	0.2111	1.50	1233.8	4.5	1234.4	10.9	1234.7	16.9

TABLE A1
(continued)

Spot	U (ppm)	Th (ppm)	Th/U	$^{206}Pb/^{204}Pb$	Pb (ppm)	f206 (%)	$^{207}Pb/^{206}Pb$	±1σ (%)	$^{207}Pb/^{235}U$	±1σ (%)	$^{206}Pb/^{238}U$	±1σ (%)	$^{207}Pb/^{206}Pb$ Age (Ma)	±1σ	$^{207}Pb/^{235}U$ Age (Ma)	± 1σ	$^{206}Pb/^{238}U$ Age (Ma)	±1σ
Sample D15JP02																		
D15JP02-01[b]	465	303	0.65	61	104.02	{30.85}	0.07221	6.28	1.82815	6.46	0.1836	1.51	991.7	122.8	1055.6	43.3	1086.8	15.1
D15JP02-02	959	1582	1.65	11459	319.73	{0.16}	0.08113	0.41	2.32946	1.56	0.2083	1.50	1224.5	8.0	1221.3	11.1	1219.5	16.7
D15JP02-03	352	296	0.84	14865	97.53	{0.13}	0.08110	0.61	2.30483	1.63	0.2061	1.51	1223.9	12.0	1213.8	11.6	1208.1	16.6
D15JP02-04	536	674	1.26	1266	162.01	{1.48}	0.08188	1.77	2.33130	2.32	0.2065	1.50	1242.6	34.4	1221.9	16.7	1210.2	16.6
D15JP02-05	719	919	1.28	1496	219.19	{1.25}	0.08032	0.74	2.29318	1.68	0.2071	1.51	1204.8	14.5	1210.2	11.9	1213.2	16.7
D15JP02-06[c]	259	165	0.64	764	65.20	{2.45}	0.07837	1.72	2.15658	2.28	0.1996	1.50	1156.2	33.8	1167.2	16.0	1173.1	16.1
D15JP02-07	961	1565	1.63	76510	319.88	{0.02}	0.08137	0.34	2.34814	1.54	0.2093	1.50	1230.5	6.7	1227.0	11.0	1225.0	16.8
D15JP02-08	337	289	0.86	33070	94.41	{0.06}	0.08023	0.59	2.31876	1.61	0.2096	1.50	1202.6	11.6	1218.0	11.5	1226.8	16.8
D15JP02-09	411	333	0.81	9427	111.67	{0.20}	0.08107	0.77	2.28717	1.69	0.2046	1.50	1223.0	15.0	1208.3	12.0	1200.1	16.5
D15JP02-10[d]	367	193	0.52	6530	91.03	{0.29}	0.08003	0.85	2.19235	1.72	0.1987	1.50	1197.8	16.6	1178.6	12.1	1168.2	16.1
D15JP02-11	364	343	0.94	1850	103.41	{1.01}	0.08233	1.37	2.33927	2.03	0.2061	1.50	1253.4	26.6	1224.3	14.6	1207.9	16.5
D15JP02-12	828	1191	1.44	40362	266.90	{0.05}	0.08100	0.38	2.37024	1.55	0.2122	1.50	1221.4	7.4	1233.7	11.1	1240.7	17.0
D15JP02-13	494	438	0.89	6837	138.78	{0.27}	0.08023	0.85	2.29659	1.73	0.2076	1.51	1202.6	16.7	1211.2	12.3	1216.1	16.7
D15JP02-14[c]	225	146	0.65	1520	56.49	{1.23}	0.08028	1.83	2.20609	2.37	0.1993	1.50	1203.8	35.6	1183.0	16.7	1171.6	16.1
D15JP02-15[b]	393	304	0.77	198	102.49	{9.46}	0.07659	4.56	2.12079	4.80	0.2008	1.50	1110.4	88.6	1155.6	33.7	1179.8	16.2
D15JP02-16[c]	257	145	0.56	1983	63.17	{0.94}	0.07954	1.15	2.14566	1.89	0.1957	1.50	1185.5	22.5	1163.7	13.2	1151.9	15.9
D15JP02-17	583	658	1.13	2123	171.89	{0.88}	0.08033	0.92	2.29313	1.76	0.2070	1.50	1205.2	18.0	1210.2	12.5	1213.0	16.6
D15JP02-18[d]	477	228	0.48	757	116.20	{2.47}	0.07898	1.20	2.17537	1.92	0.1998	1.50	1171.5	23.5	1173.2	13.4	1174.1	16.1
D15JP02-19	428	348	0.81	34017	118.29	{0.05}	0.08094	0.70	2.32724	1.66	0.2085	1.50	1220.0	13.8	1220.6	11.8	1221.0	16.7
D15JP02-20	467	499	1.07	58696	136.06	{0.03}	0.08165	0.51	2.32970	1.59	0.2069	1.50	1237.0	10.0	1221.4	11.3	1212.5	16.6
Sample D15JP03																		
D15JP03-01[c]	289	135	0.47	2000	70.01	{0.94}	0.08079	1.12	2.21430	1.87	0.1988	1.50	1216.2	21.9	1185.6	13.2	1168.8	16.1
D15JP03-02	1409	1123	0.80	870	380.30	{2.15}	0.08037	1.33	2.26622	2.01	0.2045	1.50	1206.0	26.0	1201.8	14.2	1199.5	16.5
D15JP03-03[b]	1315	1449	1.10	531	379.44	{3.52}	0.08140	1.35	2.27229	2.02	0.2025	1.50	1231.2	26.2	1203.7	14.3	1188.5	16.3
D15JP03-04[b]	679	519	0.76	147	169.27	{12.72}	0.06720	7.45	1.81835	7.60	0.1963	1.51	843.8	147.7	1052.1	51.0	1155.2	16.0
D15JP03-05[d]	2000	2389	1.19	3010	597.81	{0.62}	0.07993	0.82	2.23593	1.71	0.2029	1.50	1195.3	16.1	1192.4	12.1	1190.8	16.3
D15JP03-06	1578	1475	0.93	828	449.77	{2.26}	0.07941	1.72	2.29244	2.29	0.2094	1.50	1182.3	33.7	1210.0	16.3	1225.5	16.8
D15JP03-07[a]	3401	1346	0.40	6248	817.77	{0.30}	0.08002	0.49	2.20723	1.62	0.2001	1.55	1197.4	9.7	1183.3	11.4	1175.6	16.6
D15JP03-08	1003	945	0.94	47309	288.24	{0.04}	0.08123	0.43	2.34359	1.57	0.2092	1.50	1227.1	8.5	1225.6	11.2	1224.8	16.8
D15JP03-09[d]	2492	3879	1.56	20150	834.39	{0.09}	0.08080	0.38	2.39076	1.55	0.2146	1.50	1216.6	7.5	1239.8	11.1	1253.3	17.1
D15JP03-10[d]	1601	1325	0.83	3343	464.90	{0.56}	0.08074	0.71	2.40599	1.66	0.2161	1.50	1215.0	13.9	1244.4	12.0	1261.4	17.2
D15JP03-11	1244	1297	1.04	3949	363.48	{0.47}	0.08076	0.77	2.32414	1.69	0.2087	1.50	1215.5	15.1	1219.7	12.1	1222.0	16.7
D15JP03-12[d]	2125	2960	1.39	17945	699.53	{0.10}	0.08166	0.43	2.43325	1.56	0.2161	1.50	1237.4	8.4	1252.5	11.3	1261.3	17.2

TABLE A1
(continued)

Spot	U (ppm)	Th (ppm)	Th/U	$^{206}Pb/^{204}Pb$	Pb (ppm)	f206 (%)	$^{207}Pb/^{206}Pb$	±1σ (%)	$^{207}Pb/^{235}U$	±1σ (%)	$^{206}Pb/^{238}U$	±1σ (%)	$^{207}Pb/^{206}Pb$ Age (Ma)	±1σ	$^{207}Pb/^{235}U$ Age (Ma)	± 1σ	$^{206}Pb/^{238}U$ Age (Ma)	±1σ
Sample D15JP03																		
D15JP03-13[a]	3184	3485	1.09	5677	956.15	{0.33}	0.08085	0.39	2.31944	1.55	0.2081	1.50	1217.9	7.7	1218.3	11.1	1218.4	16.7
D15JP03-14	1307	1079	0.83	2848	365.24	{0.66}	0.08239	1.36	2.35696	2.03	0.2075	1.50	1254.8	26.4	1229.7	14.6	1215.4	16.6
D15JP03-15[b]	1898	1537	0.81	155	506.35	{12.06}	0.08124	4.17	2.26998	4.45	0.2026	1.57	1227.3	79.7	1203.0	31.9	1189.5	17.1
D15JP03-16	958	735	0.77	16398	259.98	{0.11}	0.08096	0.39	2.30691	1.55	0.2067	1.50	1220.5	7.6	1214.4	11.0	1211.0	16.6
D15JP03-17[b]	8000	5610	0.70	381	1475.01	{4.91}	0.07908	1.17	1.59165	1.91	0.1460	1.51	1174.2	22.9	966.9	12.0	878.3	12.4
D15JP03-18	1770	1654	0.93	782	496.78	{2.39}	0.08054	1.01	2.28582	1.81	0.2058	1.50	1210.3	19.8	1207.9	12.9	1206.6	16.5
D15JP03-19[b]	984	724	0.74	148	254.65	{12.64}	0.07502	5.48	2.10136	5.68	0.2031	1.50	1069.1	106.3	1149.3	39.8	1192.2	16.4
D15JP03-20	883	737	0.83	3739	243.95	{0.50}	0.07979	0.57	2.26685	1.61	0.2060	1.50	1191.9	11.2	1202.0	11.4	1207.7	16.6

Note: anomalous analysis not used for mean age calculation: a, high U content; b, high common Pb; c, little total Pb; d, the data are outliers.

TABLE A2

SIMS Pb–Pb dating results of baddeleyites from dike LC01

Spot	U (ppm)	$^{206}Pb/^{204}Pb$	Isotopic ratio		Age (Ma)	±1σ
			$^{207}Pb/^{206}Pb$	±1σ (%)	$^{207}Pb/^{206}Pb$	
D15LC01-01	505	4249	0.08119	1.24	1226.0	24.2
D15LC01-02	809	5478	0.08124	0.43	1227.2	8.4
D15LC01-03	469	480654	0.08076	0.33	1215.5	6.5
D15LC01-04	1019	105166	0.08080	0.37	1216.5	7.3
D15LC01-05	339	95031	0.08114	0.38	1225.0	7.4
D15LC01-06	1083	136935	0.08112	0.39	1224.2	7.6
D15LC01-07	986	419798	0.08055	0.39	1210.6	7.7
D15LC01-08	1191	1240	0.08018	0.88	1201.5	17.3
D15LC01-09	1484	NaN	0.08158	0.35	1235.3	6.9
D15LC01-10	771	10606	0.08073	0.43	1214.8	8.4
D15LC01-11	547	13945	0.08058	0.34	1211.2	6.7
D15LC01-12	786	7534	0.08011	0.47	1199.7	9.2
D15LC01-13	489	106312	0.08083	0.34	1217.2	6.6
D15LC01-14	638	7947	0.08072	0.50	1214.6	9.8
D15LC01-15	821	74993	0.08119	0.41	1226.0	8.0
D15LC01-16	816	6829	0.07977	0.44	1191.3	8.7
D15LC01-17[a]	3048	445509	0.08098	0.33	1220.9	6.5
D15LC01-18	497	NaN	0.08101	0.41	1221.6	8.0
D15LC01-19	494	215004	0.08139	0.35	1230.9	6.8
D15LC01-20	933	460774	0.08140	0.38	1231.0	7.5
D15LC01-21	742	45864	0.08135	0.41	1229.9	8.1
D15LC01-22	515	NaN	0.08097	0.31	1220.8	6.1
D15LC01-23	1214	9937	0.08091	0.68	1219.4	13.4
D15LC01-24	1813	11935	0.08093	0.39	1219.6	7.6
D15LC01-25	1795	474039	0.08044	0.35	1207.8	6.8

Note: anomalous analysis not used for mean age calculation: a, high U content; NaN, indicates that common lead level is very low and the test result is zero without error.

TABLE A3

Euler rotation parameters for the reconstructions

	Euler pole		Angle of Rotation	
	(°N)	(°E)	(°)	References
South Australia Craton to North Australia Craton	−18	134	51	Payne and others (2009)
West Australia Craton to North Australia Craton	−20	135	40	Li and Evans (2011)
Mawson Craton to present Australia	−2	38.9	31.5	Powell and Li (1994)
Aldan to Anabar	60	115	25	Evans (2009)
Anabar to Laurentia	77	98	137	Evans and others (2016)
~1450 Ma				
North Australia Craton to Laurentia	31.5	98	102.5	Zhang and others (2012)
Baltica to Laurentia	47.5	1.5	49	Cawood and others (2010)
NCC to Laurentia	45.7	330.8	33.4	Zhang and others (2012)
Laurentia absolute rotation	2.4	302	−110.7	This study
~1220 Ma				
Baltica to Laurentia	45.0	7.5	44.9	Cawood and others (2010)
Laurentia absolute rotation	−11	295	−95	This study
NCC absolute rotation	14.9	102.3	76.7	This study
North Australia Craton absolute rotation	−10	62	131	This study
~930 Ma				
NCC to Baltica	72.5	54.5	102.0	Zhao and others (2019)
Laurentia to Baltica	81.5	250.0	50.0	Fu and others (2015)
Baltica absolute rotation	34.4	291.0	−112.5	Zhao and others (2019)

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